

Pouya Parsa

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Research Interests: Vision-Language Models • Memory-Augmented Large Language Models

Education

Master of Science

University of Minnesota Twin Cities

Minneapolis, USA

2024–2026

- GPA: 3.7/4.00; selected coursework: Computer Vision and Generative AI.

Bachelor of Science in Computer Science

Amirkabir University of Technology (Tehran Polytechnic)

Tehran, Iran

2018–2022

- GPA: 3.72/4.00; thesis: *Transformer-Based Deep Learning Method for Portfolio Optimization*.

Publications

Visual Distribution Anchoring for Efficient Prompt Tuning

Pouya Parsa, Raoof Zare Moayed, and Seongjin Choi

arXiv:2607.28967, July 2026

- Introduced class-specific visual prototypes from an unlabeled target pool, improving mean frozen vision–language classification accuracy from 65.82% to 69.21% without target-side optimization or test-query access.

Can the Cloud Drive? Infrastructure Feasibility of Offloading Autonomous Driving Across 5G and 6G

Pouya Parsa, Kawon Han, and Seongjin Choi

arXiv:2607.09045, 2026

- Analyzed cloud offloading for vision–language–action driving models, showing when GPU memory bandwidth remains the bottleneck after 5G/6G uplink constraints are satisfied.

Video-based Vehicle Surveillance in the Wild: License Plate, Make, and Model Recognition with Self Reflective Vision-Language Models

Pouya Parsa, Keya Li, Kara M. Kockelman, and Seongjin Choi

arXiv:2508.01387; TRB 2026

- Developed a video vision–language pipeline achieving 91.67% license-plate and 66.67% make-and-model top-1 accuracy on smartphone data; self-reflection improved make-and-model results by 5.72% on average.

Where2Start: Leveraging Initial States for Robust and Sample-Efficient Reinforcement Learning

Pouya Parsa, Raoof Zare Moayed, Mohammad Bornosi, and Mohammad Mahdi Bejani

arXiv:2311.15089, 2023

- Proposed an informative initial-state selection method that achieved up to an $8\times$ improvement in sample efficiency over reinforcement-learning baselines.

Experience

Research Assistant

University of Minnesota; advisor: Dr. Seongjin Choi

Minneapolis, USA

Aug 2024–Present

- Conduct research in machine learning with an emphasis on vision–language models, computer vision, and generative AI.
- Develop methods for self-reflective recognition, efficient prompt tuning, and infrastructure-aware autonomous-driving inference.

Project Manager

XTON

Remote (Dubai)

Apr 2024–Jul 2024

- Managed infrastructure for a high-availability crypto clicker application serving more than 500,000 users.
- Implemented a serverless architecture using AWS Lambda and auto-scaling and coordinated stakeholder expectations.

Machine Learning Engineer

ZebraCat.ai

Remote (Germany)

Sep 2022–Mar 2023

- Developed embedding-based video retrieval and semantic video-segmentation methods.
- Collected a dataset of more than one million advertisement videos.

Data Science Intern

MCI (Hamrahe Aval)

Tehran, Iran

Jul 2022–Sep 2022

- Built a fault-tolerant ETL pipeline and collaborative-filtering service for recommender systems at Iran's largest mobile operator, delivering recommendations in under 10 ms.

Machine Learning Engineer Intern

University of Zurich

Zurich, Switzerland

Jan 2022–Mar 2022

- Developed interfaces for data ingestion, preprocessing, machine learning, visualization, and report generation.

Honors and Awards

Departmental Fellowship Award

University of Minnesota Twin Cities

2026

Awarded in recognition of academic excellence and research potential.

CTS TRB Reimbursement Award

Center for Transportation Studies

2026

Received competitive travel reimbursement to present research at TRB 2026.

Outstanding Student Recognition

Amirkabir University of Technology Honors and Olympiads Program

Recognized for academic excellence; ranked 403rd among approximately 150,000 participants in Iran's national university entrance examination.

Technical Skills

Programming: Python, Java, C/C++

Machine Learning: PyTorch, TensorFlow, scikit-learn, OpenCV

Data: NumPy, pandas, SQL, NoSQL

Selected Projects

Heart Rate Measurement from Video

Estimated heart rate non-invasively from subtle color changes in facial video.

[Project link](#)

Neural-Network Optimizer Comparison

Compared particle swarm optimization with Adam for neural-network training.

[Project link](#)

Movie Recommender

Implemented collaborative filtering with singular value decomposition.

[Project link](#)

Teaching and Service

Teaching Assistant: Computer Vision (Winter 2023); Optimization in Neural Networks (Winter 2023); Artificial Intelligence (Winter 2022).

Service: Board Member, Scientific Association, Amirkabir University of Technology (2020–2021); Member, Khayam–Turing Machine Learning and Data Science Competition Group (2020–2021).

Languages

English: Fluent (TOEFL iBT 104/120) **Persian:** Native